Phase D4 Key Prior-Art Claims Index

Derivation Note D4.29 — #634 in the CKS Defensive-Publication Series

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

Phase D4 of the CKS derivation note series comprises twenty-six composition pair notes (D4.02–D4.27) and two framing notes. Each composition pair note demonstrates, as prior art, that a specific governance scenario combining two architectural commitments simultaneously is not novel. This note, D4.29, functions as the navigation index for that body of work. It collects the twenty-five key prior-art claims established by Phase D4, states each claim concisely, identifies the establishing composition pair note, and groups claims into four thematic clusters to support efficient navigation. The index is the instrument that converts Phase D4 from a reading project into a searchable prior-art reference. D4.30 follows as Phase D4 final closure.

1. Why a key claims index is needed

The defensive-publication function of Phase D4 depends on accessibility. Twenty-six composition pair notes, read sequentially, demonstrate that the space of dual-commitment governance scenarios is occupied prior art. But sequential reading is not how prior art is used in practice. An adversary claiming novelty for "governed AI coordination combining X and Y" prompts a specific question: does the prior art address this particular combination? The answer requires a lookup mechanism, not a reading assignment.

The key claims index provides that mechanism. Each of the twenty-five claims in this index forecloses a specific adversarial argument—the argument that a governance architecture combining two specified commitments simultaneously is novel. The claim does not assert that the scenario is trivial or unimportant; it asserts that the scenario was established in full, with both commitments engaged, before any third party's filing or disclosure. The establishing pair reference points to the note where that full treatment can be found.

Three properties make the index usable as a prior-art navigation tool. First, each claim is stated affirmatively—a direct anticipation argument, which is stronger than an obviousness argument and does not require the reader to supply a reasoning chain. Second, the establishing pair reference is explicit, pointing to the full governance treatment. Third, the thematic cluster grouping in §§3–6 enables navigation by adversarial attack type rather than by note number.

2. How to use this index

To use D4.29 as a prior-art navigation tool: (1) Identify the adversarial claim's subject area. (2) Go to the corresponding cluster in §§3–6. (3) Locate the claim that matches the combination of commitments in the adversarial claim. (4) Use the establishing pair reference to navigate to the full treatment. When an adversarial claim spans multiple clusters, claims from each relevant cluster can be combined as independent grounds for anticipation.

3. Cluster 1 — Governance Integrity

Claims in this cluster are most relevant when defending against adversarial arguments that governance under scenario pressure—emergency conditions, high frequency, speed requirements, portfolio-scale exits, or governance health degradation—is a novel design problem.

Claim D4-01.

Pre-authorization resolves the speed-governance tension: a governance floor and rapid execution are simultaneously achievable only through pre-authorization. The composition of speed requirements with governance floor commitments is disclosed. *(Established by D4.02)*

Claim D4-04.

Exit rights exercise triggers emergency dissolution for active events—exit does not mean abandonment; dissolution governance continues to apply through the exit event. The composition of exit rights with dissolution governance is disclosed. *(Established by D4.05)*

Claim D4-05.

Governance health thresholds must be frequency-adjusted: at high frequency, absolute thresholds are misleading and relative thresholds are required. The composition of high-frequency operation with health threshold governance is disclosed. *(Established by D4.06)*

Claim D4-13.

Emergency dissolution requires pre-authorized evolution feed configuration—emergency conditions do not suspend evolution feed governance. The composition of emergency dissolution with evolution feed governance is disclosed. *(Established by D4.14)*

Claim D4-17.

In portfolio-scale exit, participation configuration update must precede relationship-specific exit execution to prevent new relationships forming during exit. The composition of portfolio-scale exit with participation configuration governance is disclosed. *(Established by D4.18)*

Claim D4-23.

Governance-traceable determinism—determinism traceable to authored governance content—is distinct from computational determinism; pre-authorized labor satisfies both commitments simultaneously. The composition of determinism commitments with governance authorship is disclosed. *(Established by D4.25 (with D4.24 contributing the theoretical foundation))*

Claim D4-24.

Governance health indicator warning thresholds must include post-mortem trigger criteria; the calibration feedback loop connects post-mortem findings to indicator calibration. The composition of health monitoring with post-mortem governance is disclosed. *(Established by D4.26)*

Claim D4-25.

The minimum viable governance floor is the testable definition of non-specialist governance accessibility; each floor requirement must be decomposable into authority and labor dimensions. The composition of governance accessibility commitments with floor decomposition is disclosed. *(Established by D4.27)*

4. Cluster 2 — Relationship Architecture

Claims in this cluster are most relevant when defending against adversarial arguments that the governance structure of sustained multi-party coordination relationships—including cardinality configurations, parent-child document hierarchies, asymmetric capacity calibration, and multi-domain portfolio management—is novel.

Claim D4-02.

Bilateral standing configurations do not automatically compose into N-ary governance—N>2 events require either multilateral configuration or bilateral configurations plus an N-specific supplement. The composition of bilateral governance structures with N-ary cardinality requirements is disclosed. *(Established by D4.03)*

Claim D4-03.

Competition variant selection and preserve-tier-dominant conflict routing are separate configured objects that must both be explicitly authored. The composition of competition configuration with conflict routing governance is disclosed. *(Established by D4.04)*

Claim D4-06.

Governance capacity is an input to participation configuration, not merely a constraint on its outputs. The composition of capacity assessment with participation governance design is disclosed. *(Established by D4.07)*

Claim D4-08.

Conflict carry-through annotation entry tier determines propagation path through the vertical evolution hierarchy. The composition of conflict annotation with evolution hierarchy routing is disclosed. *(Established by D4.09)*

Claim D4-12.

Cross-organizational agreement is the parent governance document; standing configurations are child documents; the parent governs inconsistencies; agreement termination revokes all standing configurations. The composition of parent-child document governance with agreement termination effects is disclosed. *(Established by D4.13)*

Claim D4-14.

Partial withdrawal creates the WITHDRAWN-SIDE conflict class requiring status update, routing decision, and post-withdrawal conflict detection. The composition of partial withdrawal with conflict class governance is disclosed. *(Established by D4.15)*

Claim D4-15.

In asymmetric coordination relationships, standing configuration complexity must be calibrated to the simpler party's governance capacity. The composition of asymmetric relationship governance with capacity calibration is disclosed. *(Established by D4.16)*

Claim D4-16.

High-frequency coordination relationships require four specific agreement provisions: acknowledged governance practices, escalation overflow, amendment velocity, and health remediation clause. The composition of high-frequency operation with sustained relationship agreement governance is disclosed. *(Established by D4.17)*

Claim D4-22.

Multi-domain parties require domain-specific agreement portfolios; cross-domain contamination must be prevented through portfolio governance. The composition of multi-domain participation with agreement portfolio architecture is disclosed. *(Established by D4.23)*

5. Cluster 3 — Knowledge and Learning

Claims in this cluster are most relevant when defending against adversarial arguments that the governance of knowledge transfer, absorption sequencing, trust-calibrated review, and intelligence extraction from coordination events is novel.

Claim D4-07.

Knowledge absorption must be completed for a target aspect before that aspect undergoes restructuring. The composition of absorption sequencing with restructuring governance is disclosed. *(Established by D4.08)*

Claim D4-10.

In knowledge transfer context, absorption intent must not become absorption obligation—the cross-organizational agreement must explicitly state this distinction. The composition of intent governance with cross-organizational agreement drafting is disclosed. *(Established by D4.11)*

Claim D4-11.

For competitive intelligence events, post-mortem review is the intelligence extraction mechanism; three-way conflict pattern analysis is required. The composition of competitive intelligence governance with post-mortem extraction is disclosed. *(Established by D4.12)*

Claim D4-18.

Trust calibration affects absorption review depth, not the review requirement; absorption outcomes feed back into trust calibration. The composition of trust calibration with absorption review governance is disclosed. *(Established by D4.19)*

6. Cluster 4 — Documentation and Perimeter

Claims in this cluster are most relevant when defending against adversarial arguments that the governance of audit records, retention policies, perimeter defense sequencing, version history standards, and research record categories is novel.

Claim D4-09.

When both regulatory audit requirements and documentation standards apply, the higher of the two requirements governs each dimension—a dual floor. The composition of regulatory audit governance with documentation standard governance is disclosed. *(Established by D4.10)*

Claim D4-19.

Observational study coordination events require dual retention policies and explicit consent linkage in governance records; research record is an eighteenth record category. The composition of research study governance with retention policy architecture is disclosed. *(Established by D4.20)*

Claim D4-20.

Contribution scope is the first line of perimeter defense; integrity checks are the second line; checks must be calibrated to contribution scope. The composition of contribution scope governance with integrity check calibration is disclosed. *(Established by D4.21)*

Claim D4-21.

Dispute-evidence-ready version history requires prior values, specific authority linkage, and mutual accessibility. The composition of version history governance with dispute-evidence readiness is disclosed. *(Established by D4.22)*

7. Summary table

Claim	Short description	Cluster	Est. by
D4-01	Pre-authorization resolves speed-governance tension	Gov. Integrity	D4.02
D4-02	Bilateral configs do not auto-compose into N-ary	Rel. Architecture	D4.03
D4-03	Competition variant selection and preserve-tier routing are separate objects	Rel. Architecture	D4.04
D4-04	Exit rights trigger emergency dissolution	Gov. Integrity	D4.05
D4-05	Health thresholds must be frequency-adjusted	Gov. Integrity	D4.06
D4-06	Governance capacity is input to participation configuration	Rel. Architecture	D4.07
D4-07	Absorption completed before restructuring	Knowledge & Learning	D4.08
D4-08	Conflict annotation tier determines propagation path	Rel. Architecture	D4.09
D4-09	Dual floor when audit and documentation standards apply	Doc. & Perimeter	D4.10
D4-10	Absorption intent must not become obligation	Knowledge & Learning	D4.11
D4-11	Post-mortem is intelligence extraction mechanism	Knowledge & Learning	D4.12
D4-12	Cross-org agreement is parent; standing configs are children	Rel. Architecture	D4.13

D4-13	Emergency dissolution requires pre-authorized evolution feed	Gov. Integrity	D4.14
D4-14	Partial withdrawal creates WITHDRAWN-SIDE conflict class	Rel. Architecture	D4.15
D4-15	Asymmetric relationship calibrated to simpler capacity	Rel. Architecture	D4.16
D4-16	High-frequency relationships require four provisions	Rel. Architecture	D4.17
D4-17	Portfolio-scale exit: config update precedes execution	Gov. Integrity	D4.18
D4-18	Trust calibration affects absorption review depth	Knowledge & Learning	D4.19
D4-19	Observational study: dual retention and consent linkage	Doc. & Perimeter	D4.20
D4-20	Contribution scope is first perimeter defense line	Doc. & Perimeter	D4.21
D4-21	Dispute-evidence-ready version history requirements	Doc. & Perimeter	D4.22
D4-22	Multi-domain parties require domain-specific portfolios	Rel. Architecture	D4.23
D4-23	Gov.-traceable determinism distinct from computational	Gov. Integrity	D4.25
D4-24	Health warning thresholds include post-mortem triggers	Gov. Integrity	D4.26
D4-25	Minimum viable governance floor is testable definition	Gov. Integrity	D4.27

8. Conclusion and what follows

The twenty-five claims collected here represent the affirmative prior-art output of Phase D4. Together they establish that the space of dual-commitment governance scenarios at the inter-Self coordination scope—spanning governance integrity under pressure, multi-party relationship architecture, knowledge transfer and learning, and documentation and perimeter defense—was occupied in full before any third party's filing or disclosure. No single claim is the most important; the coverage is the contribution. A sufficiently comprehensive space of two-commitment combinations leaves no room for adversarial novelty arguments that rely on combining two established commitments.

The index is designed to remain usable as the surrounding literature evolves. Each claim is stated without jargon that ties it to a specific implementation context, so that the mapping from an adversarial claim's vocabulary to a Phase D4 claim is a semantic question, not a terminological one. The reader who finds a near-match but not an exact match should read the full establishing pair note; the full treatment discloses the commitment space rather than only the specific claim label.

D4.30 follows as Phase D4 final closure, completing the Series D composition pair phase.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model.* April 2026. ORCID: 0009-0004-8065-3235.

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